Towards AI-Assisted Pituitary Surgery – The Systematic Translation of Surgical Workflow Analysis to Improve Training, Audit and Prognostication.
NIHR Doctoral Fellowship Application

Summary
This proposal leverages artificial intelligence (AI) to address key unmet surgical needs, with the ultimate goal of improving patient outcomes, using pituitary surgery as an illustrative exemplar of its potential.

Pituitary tumours are amongst the most common brain tumours, and may present in a variety of ways, including visual decline and hormone imbalance. Left untreated, they may result in significant systemic morbidity, mortality and quality of life impairment. Transsphenoidal surgery is the gold standard treatment for most patients with symptomatic functioning and non-functioning pituitary adenomas. However, this is a challenging operation to learn and perform, and requires years of dedicated training in specialist pituitary centres. Different surgeons perform these surgeries in distinct ways, with variance in the recording of surgical performance and resultant outcomes. Therefore, auditing surgical practice, which is critical for safe service delivery, is challenging. The variation in surgical performance, along with patient differences, also makes the prediction of post-operative outcomes very difficult. Consequently, it is challenging to tailor post-operative care to each patient, for example recognising patients who may be at low risk for complications and safely discharged earlier, or those at higher risk of complications who may need close monitoring and early intervention.

The first step to tackling these problems is to understand how these surgeries are performed and compare the relationships between their performance and outcomes. Operative workflow analysis enables this by breaking down operations into their fundamental building blocks, including the surgical steps performed and the instruments used. Practically speaking this analysis is performed by segmenting surgical videos. This is very labour-intensive if undertaken manually but artificial intelligence (AI) has the potential to perform this analysis automatically, accurately and at a large scale.

We have developed pre-clinical AI models capable of recognising surgical workflow in endoscopic pituitary surgery. The next step of this work is to apply this technology to the unmet clinical needs of pituitary surgery, and I plan to do this over a 3-year doctoral fellowship, guided by the IDEAL and DECIDE-AI guidelines, with 3 aims:

1) To investigate the use of AI workflow analysis to **improve audit** by assisting surgeons in creating semi-automated operative notes. We hypothesize these will be more accurate and comprehensive, whilst helping relieve some administrative burden.
2) To examine the uses of AI workflow analysis to **address the training needs** of surgical teams, by presenting a real-time analysis of current and upcoming surgical steps and instruments, with surgical performance metrics. We hypothesize this will improve the educational yield from each case by orientating the team during the procedure and by facilitating coaching by mentors.
3) To explore the use of AI workflow analysis for **prognostication** by linking this analysis with clinical outcomes, allowing data-driven post-operative outcome prediction. We hypothesize this will allow more patient-specific post-operative management.

I plan to translate each of these technologies via pilot studies (examining feasibility, acceptability and usability) and single centre development studies (examining effectiveness and safety).

We have an established team of neurosurgeons (based at the UK's largest pituitary centre), engineers (based at a world-leading surgical innovation institute) and key industry collaborators (Medtronic and NVIDIA). Having built a secure data pipeline, generated a large dataset, and created foundational preclinical models through an NIHR Academic Clinical Fellowship and a Cancer Research UK Predoctoral Fellowship, I am confident I can deliver this proposal.

Vision

Pituitary adenomas are among the most common intracranial tumours, with an estimated prevalence of up to 20%[1, 2]. They may present incidentally, through mass effect (e.g. visual decline) or hormone imbalance (e.g. acromegaly), therefore potentially causing significant morbidity, quality of life reduction and death if left untreated[1, 3]. The gold standard treatment for most patients with symptomatic pituitary adenoma is transsphenoidal surgical excision, with the endoscopic transsphenoidal approach (eTSA) bolstering access to these tumours[4].

However, these operations are technically demanding with long learning curves, posing a training challenge for surgeons, and therefore requiring post-training fellowships to attain the necessary clinical and technical skills[5-8]. Furthermore, there is variation in how these operations are performed (largely based on surgeon preference), which exacerbates this training challenge and may result in differing outcomes[3, 9, 10]. Through executing a systematic review and a large multicentre observational study, I have found significant heterogeneity in the closure phase alone - in how it was performed, how operative details were recorded and how it affected outcomes (i.e. whether it prevented the leakage of cerebrospinal fluid [CSF])[11-15]. This heterogeneity in practice and operative documentation poses a significant challenge to the clinical audit of performances and maintaining safety. Similarly, the resulting wide variations in outcomes, dependent on both operative and patient factors, make post-operative prognostication notoriously difficult[15-17]. Therefore, patients are often monitored for 3-5 days to monitor for common complications (e.g. sodium imbalance), and initiatives to facilitate earlier discharge or prophylactic treatments (e.g. fluid restriction) in selected cases have not become widespread[15-17]. The inability to robustly predict outcomes and stratify patients poses a fundamental barrier to these initiatives[16, 17].

To begin to address these unmet audit, training and clinical effectiveness needs, we generated an international, consensus-driven analysis of the operative workflow (surgical steps, instruments and errors) in contemporary pituitary surgery[18-20]. This workflow analysis is used to systematically codify operations into their fundamental components, to allow in-depth analysis and identification of operative variations[18]. This can be performed through the segmentation of operative videos. However, done manually, this is extremely labour-intensive - posing limitations to its upscaling and analysis. Recent advances in artificial intelligence (AI), more specifically machine learning and computer vision, have allowed this operative workflow analysis to be generated automatically[19, 21, 22]. Thus, we have developed novel pre-clinical AI models capable of accurately and automatically recognising pituitary surgery workflow from operative videos – a world first for pituitary surgery[23-26].

The next step of this work is to apply this technology to the unmet clinical needs of pituitary surgery[19]. I plan to do this over a 3-year doctoral fellowship, performing earlier clinical translation guided by the IDEAL and DECIDE-AI guidelines[27-30], with 3 aims:

1) To investigate the use of AI workflow analysis to **improve audit** by assisting surgeons in creating semi-automated operative notes[25]. We hypothesize these will be more accurate and comprehensive, whilst helping relieve some administrative burden.
2) To examine the uses of AI workflow analysis to **address the training needs** of surgical teams, by presenting a real-time analysis of current and upcoming surgical steps and instruments, with surgical performance metrics. We hypothesize this will improve the educational yield from each case by orientating the team during the procedure and by facilitating coaching by mentors[31, 32].
3) To explore the use of AI workflow analysis for **prognostication** by linking this analysis with clinical outcomes, allowing data-driven post-operative outcome prediction. We hypothesize this will allow more patient-specific post-operative management (e.g. reducing the length of stay or improving complication recognition)[19].

In this proposal, pituitary surgery serves as an illustrative exemplar for exploring the potential benefits of this novel technology in surgery, with the ultimate goal of improving patient outcomes. Our findings could inform improvements in the audit, training and outcomes of any type of surgery, once a data pipeline is established. The potential for this work is therefore vast, dovetailing with the MRC's remit for *innovative translational research*, the MRC *Vision for AI*, and the innovation objectives of the MRC *Strategic Delivery Plan* (2022-2025).

Approach

My vision of translating AI-powered surgical workflow analysis to improve audit, training and prognostication, using pituitary surgery as an exemplar, builds on foundations laid by previous funded projects. This includes an NIHR Academic Clinical Fellowship (2021-2024) and a Cancer Research UK Predoctoral Fellowship (awarded March 2022, for 20,000gbp over 1 year). These programmes have allowed me to collect pilot data (annotated videos and associated clinical data) for 100 pituitary adenoma patients, the largest dataset of its kind. They have also supported the building of a pipeline for secure data collection (including regional ethics committee approval and industry collaboration), and the development of novel preclinical AI models capable of accurately and automatically recognising operative workflow[23-26].

My plan to systematically clinically translate this surgical video workflow AI technology is via three parallel work packages (WP) over three years (Figures 1 & 2). This plan is guided by the IDEAL framework for evaluating surgical innovation and DECIDE AI guidelines for early clinical studies of AI interventions, with the ultimate goal of readying this technology for subsequent multicentre comparative studies[27-30]. The WPs are listed in order of projected implementation difficulty (from easy to more challenging) and will be staggered as such (Figure 2). WP1 could overlap with WP2 and WP3 if needed, but WP2 and WP3 will not overlap as they may confound each other's analysis. I have chosen multiple WPs to de-risk this application, and increase the likelihood of eventually translating at least one WP to clinical practice in the future.

WP1: Applying AI workflow analysis to assist surgical audit.

The goal of this stage is to apply our pre-clinical AI model, which uses AI workflow analysis to recognise the steps performed during an operation and populate an operative note template, to improve the accuracy, efficiency and standardisation of our surgical documentation[25].

The next stage of this work is a first-in-human pilot study (IDEAL Stage 1) of the feasibility, acceptability and accuracy of AI-assisted operative notes, where the surgeon is presented with an AI-generated note immediately post-operatively (based on the operative video). This note will be copied and pasted into our EMR after the surgeon's editing to ensure representativeness and add context (i.e. information not possible to glean from video alone). Six prospective cases will used, recording the feasibility (number of AI-assisted notes completed, time taken) and acceptability (via structured questionnaire) as primary outcomes[27, 33]. Additionally, accuracy, comprehensiveness and clarity will be assessed via an expert surgeon of the video and operative note, using questionnaires based on our consensus framework for core surgical steps and the Royal College of Surgeon's guidance for operation documentation[18, 34]. Human factors analysis will be gathered via observation and interview, and used to iterate the AI output interface and usability[27].

We will then perform a single centre prospective development study (IDEAL Stage 2a) of 30 prospective cases, comparing accuracy, comprehensiveness and clarity to a retrospective cohort of 30 operative notes and videos with no AI documentation support. Again, the AI-human interface will be continually improved with iterative human factors analysis. The Reproducibility section details outcome metrics in further detail. This study will provide the necessary experience and technological refinement for planning multicentre explorative (IDEAL Stage 2b) and comparative (IDEAL Stage 3) studies, which we plan to pursue via a dedicated project grant in the future.

WP2: Applying real-time AI workflow analysis for assisting training.

The goal of this stage is to present workflow analysis (surgical steps and instruments) to surgical teams in real-time during an operation and post-operatively to promote reflective practice, and to assess its educational and operational benefits. Our existing ethics approval (IRAS 127474) will facilitate this work. We have already performed pre-clinical (IDEAL Stage 0) studies of the candidate AI model, which is now capable of accurate and automatic surgical phase (F1 score 90%) and step (F1 score 75%) recognition (see Reproducibility section)[23, 24, 26].

The next stage of this work is a first-in-human pilot study (IDEAL Stage 1) of the feasibility, acceptability and safety of real-time intra-operative workflow analysis[27, 33]. This would be single centre, recruiting six prospective cases[30]. We will deploy our AI model via the NVIDIA Clara AGX™ system which will receive live surgical video from the surgical endoscope, and display on a second screen the current and upcoming surgical steps and associated instruments. This will be displayed to the surgical assistants, scrub nurses and the wider surgical team but not to the primary operating surgeon. The operative video will be available to the surgeons immediately post-

operatively (via the Touch Surgery™ Enterprise online platform), automatically indexed for surgical workflow, with performance metrics available for review (e.g. step frequency, order and duration). Performing surgeons will be presented with this annotated video and associated performance metrics, and guided by expert surgeons to analyse the operative strengths and areas to improve for each case – akin to performance coaching in other high-stakes disciplines[35]. The outcomes of this study will be assessing feasibility (functioning display of intra-op workflow and availability of video metrics post-op), acceptability and perceived educational yield of intra and post-operative workflow display[27, 30, 35]. Safety concerns (e.g. distraction of surgical teams by the AI display) will be actively monitored for and assessed via questionnaires[27]. We will also conduct a human factors analysis to improve how the AI model outcome is displayed via observation and interviews[27]. Data will be collected on a case-by-case basis, guided by the DECIDE-AI guidelines, and used to iteratively improve the AI-human interface[27].

We will then perform a single centre prospective development study (IDEAL Stage 2a) to assess the potential effectiveness and safety of this AI workflow display in 30 prospective cases[27, 30]. The primary outcomes of this study will be assessing the perceived educational value of seeing current and upcoming surgical steps and instruments for all members of the surgical team, and the perceived educational value of the post-operative video-assisted coaching[35]. Both will be assessed via structured questionnaire[35]. We will also assess for changes in operative time, performance and outcomes when compared to a similar retrospective cohort. Outcomes will be recorded via an established prospectively maintained database[36]. These post-operative outcome metrics will be derived from our previous multicentre collaborative pituitary surgery outcomes study[12-15]. Operative performance will be assessed using independent modified OSAT ratings of each surgical video[37]. Human factors analysis and safety monitoring will continue and iteratively improve the technology interface, again guided by the DECIDE-AI guidelines[27, 30]. Outcomes metrics for all stages are detailed further in the Reproducibility section. This study will provide the necessary experience and technological refinement for planning multicentre explorative (IDEAL Stage 2b) and comparative (IDEAL Stage 3) studies, which we plan to pursue via a dedicated project grant in the future[27, 30].

WP3: Applying AI workflow analysis to assist surgical outcome prediction & improve clinical effectiveness.

The codification of surgical videos using workflow analysis enables outcome prediction for each surgery, which can be presented to clinicians post-operatively as a decision-support tool[19, 38-40]. Supported by a previous CRUK predoctoral fellowship and an upcoming EPSRC project grant (see Related Funding section), we have developed and are refining pre-clinical AI models capable of outcome risk profiling through operative video workflow analysis.

The first step in clinical translation is a single centre first-in-human pilot study (IDEAL Stage 1) of six prospective cases[27, 30]**.** After input of the operative video data, the predicted outcome profile will be displayed (via NVIDIA Clara AGX™) to expert surgeons, trainee surgeons and specialist nurses. As the ultimate goal of this technology in the first instance is to support decision-making post-operatively but pre-discharge, only common inpatient outcomes that were readily quantifiable were included (see Reproducibility section)[14, 36]. We will explore the perceived acceptability, feasibility and utility of the AI display, as well as qualitative feedback guided by human factors analysis[27, 33]. This will be used to iteratively improve the AI-surgeon display interface[27, 30].

Next, a single centre prospective development study (IDEAL Stage 2a) will be performed to assess the potential effectiveness and safety of the AI model display in 30 prospective cases when displayed to expert surgeons, trainee surgeons and specialist nurses looking after the patient[27, 30]. AI model outputs will be made available after the operation and for the duration of the patient's stay. At discharge, each participant will be asked, via structured questionnaire, about the technology's perceived effectiveness (whether it assisted with earlier recognition or prophylactic treatment of complications for high-risk cases, or whether it facilitated earlier safe discharge for low-risk cases) and perceived safety (via human factors orientated interviews)[27]. This study will provide the necessary experience and technological refinement for planning multicentre explorative (IDEAL Stage 2b) and comparative (IDEAL Stage 3) studies, which we plan to pursue via a dedicated project grant in the future[27, 30].

Research environment

This proposal is enabled by the unique interdisciplinary environment at the Wellcome/EPSRC

Centre for Interventional & Surgical Sciences (WEISS) at University College London (UCL), UCL Neuroscience, and the National Hospital for Neurology and Neurosurgery (NHNN). UCL Neuroscience is currently rated first in Europe and second in the world by ISI Essential Science Indicators. This neuroscience expertise dovetails with the established AI and clinical innovation expertise at the Wellcome/EPSRC Centre for Interventional & Surgical Sciences (WEISS). With this interdisciplinary synergy, UCL brings together experts in the fields of surgical and interventional sciences across medical and engineering university departments. Furthermore, NHNN is part of the largest pituitary centre in the UK and it alone delivers ~150 pituitary surgeries per year. The patient base it serves is international and diverse in terms of background, therefore we will ensure our datasets and models incorporate this diversity, to improve the generalisability of our output. Altogether, the integration of these centres provides an exciting opportunity for bi-directional bedside-bench translation, and the infrastructure for this proposal to flourish.

By the start of this proposed fellowship I would have been working at WEISS, UCL and NHNN for four years – one as a clinical research fellow, followed by three years as an NIHR academic clinical fellow. During this time, I have dedicated my research time to the necessary foundational work needed for this PhD proposal. Furthermore, the last six months of my ACF will be research time spent continuing my ACF projects, which are precursors to my proposed fellowship (Figure 2). Throughout these four years, I have been mentored by my proposed PhD supervisors, and have established relationships with key collaborators (e.g. IDEAL collaboration) and industry partners (Medtronic and NVIDIA).

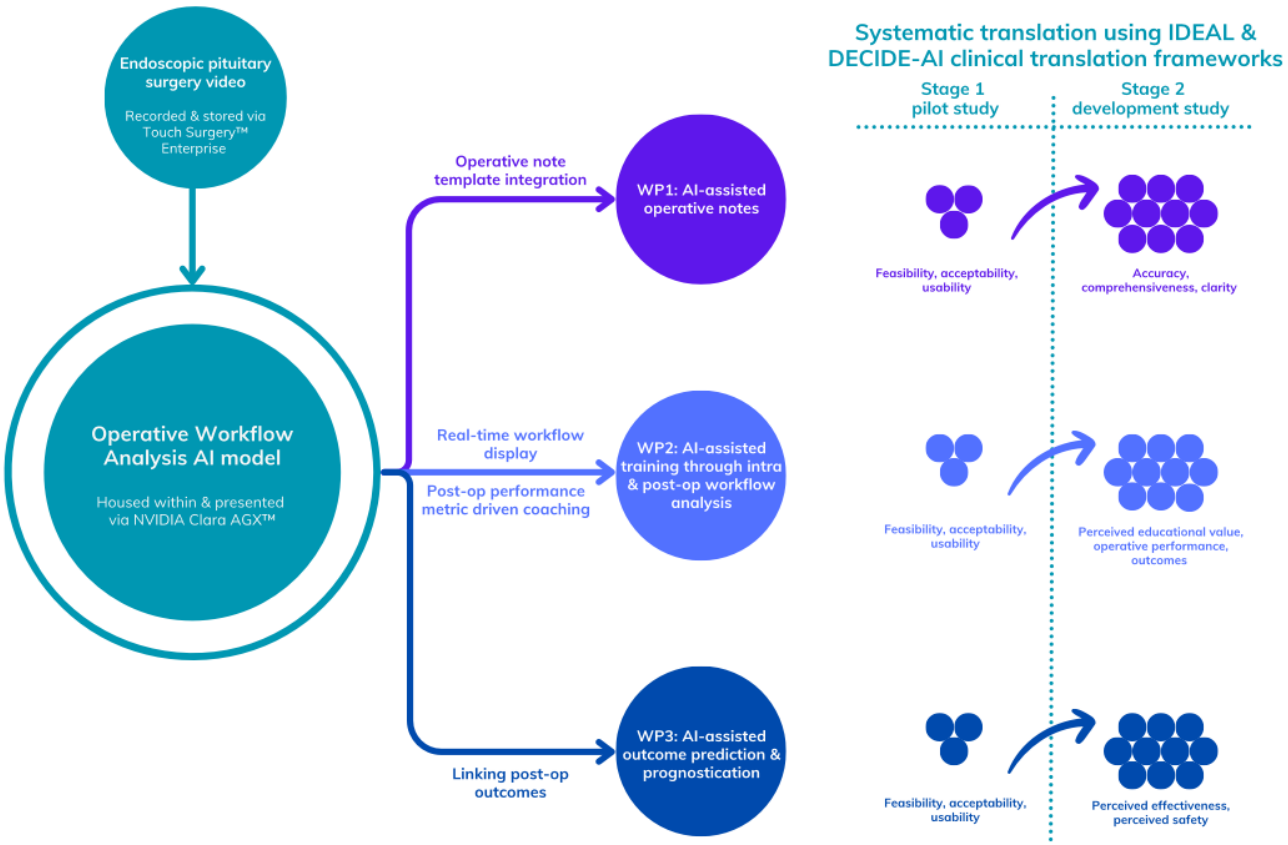

Figure 1: Proposal overview, with three parallel work packages stemming from pre-clinical AI-powered workflow analysis models, and study outcomes per stage.

Diagrammatic Workplan

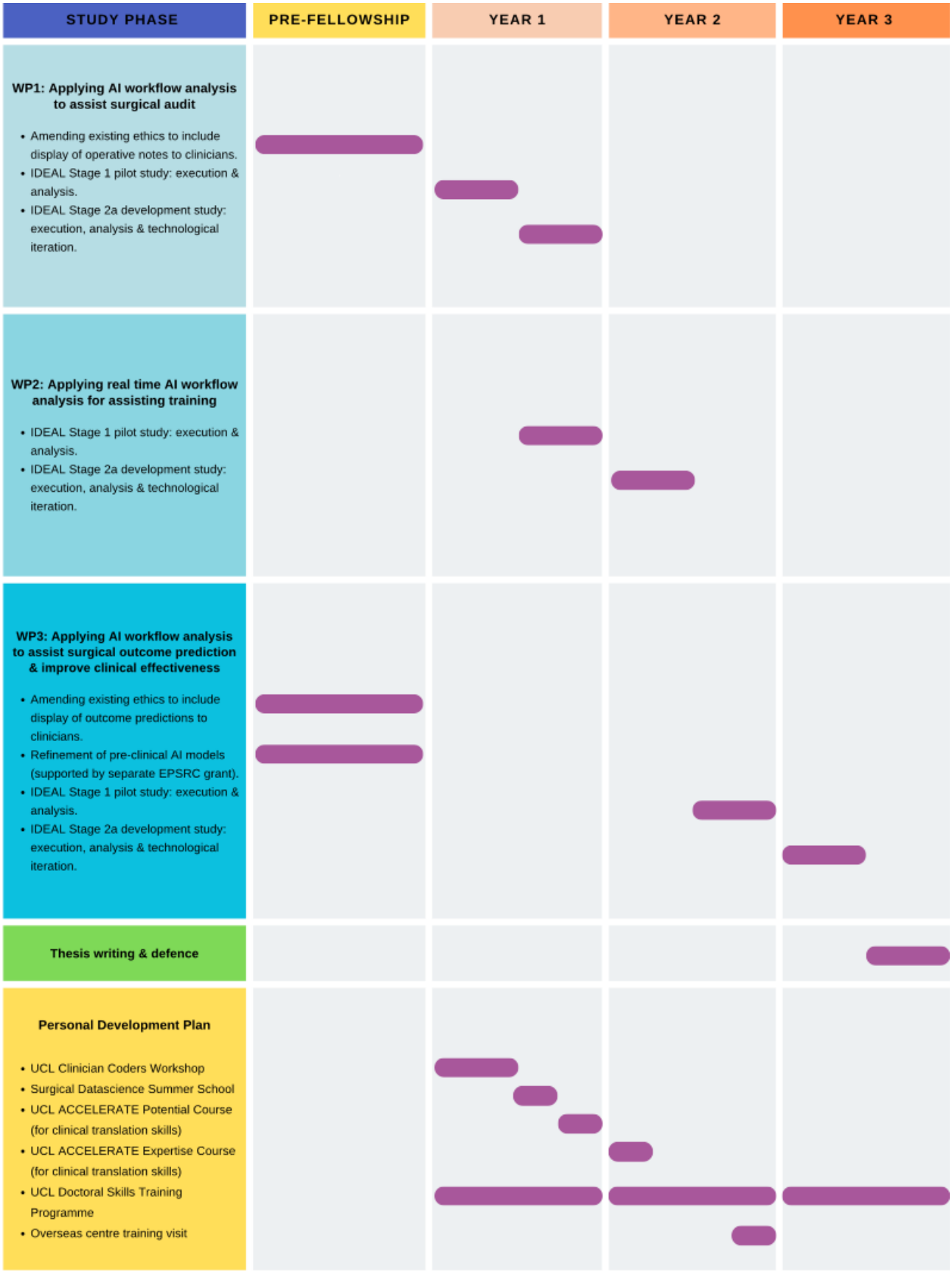

Figure 2: Gantt chart display of project stages and milestones. The pre-fellowship period represents the last six months of my ACF, which will be protected research time spent continuing my ACF projects, which are precursors to my proposed fellowship. Our centre performs 150 pituitary operations per year (therefore 12-13 per month) so patient recruitment is not an anticipated issue. However, an average of six months per study has been assigned to allow iterative improvements in the AI interface, and give buffers for implementation.

Reproducibility

Existing AI model architecture from previous IDEAL Stage 0 studies

After informed consent, each surgical video was uploaded to Touch Surgery™ Enterprise and annotated for surgical steps by expert surgeons[18, 23-26]. The resulting annotated video database (n=100 videos), was split into training and testing datasets (80:20 ratio, 5-fold stratified cross-validation). The videos were sampled at 1 frame per second, serving as the visual input for a two-stage artificial neural network (ANN) which aimed to predict the step shown in a given frame. In the first stage, a Convolutional Neural Network (CNN) was pre-trained to recognize steps and phases from a short temporal window (1-5 frames). From a computer vision perspective, due to the ambiguity of the different anatomical and instrument landmarks visible in the dynamic field of view during different steps, a single frame or short sequence may not carry sufficient information to aid correct classification[23]. To compensate for this, a second-stage Recurrent Neural Network (RNN) was trained to improve temporal resolution and prediction consistency. Step recognition was primarily evaluated using F1 scores (the harmonic mean between precision and recall), calculated per class and then summed across classes, weighted to the prevalence of the respective class. The baseline model achieved a 75% F1 score for step recognition[23], which is continually improving with increasing data input and model refinement. This model is used in WP2 to recognise intra-operative workflow in real-time and generate performance analytics, for example, surgical step order, duration and frequency. For the operative notes generation (WP1), an additional post-processing stage was implemented to ensure the relationships between steps are clinically consistent, and allow for integration with an operative note template[25]. For the outcome prediction model (WP2), a feed-forward ANN will be developed, taking the output of the existing CNN-RNN model as an input, in addition to post-operative data, to predict the outcome probabilities for each patient. All ANNs will be trained using an appropriate backpropagation method (e.g. adaptive gradient descent) to minimise the appropriate loss function (e.g. cross entropy). Hyperparameters (including learning rate, and activation function) will be optimised[41]. Final model performance will be compared against logistic regression, with averaged receiver operating characteristic curves created using 10-fold stratified cross-validation. A probability cut-off point of 0.5 will classify the predicted outcome as low or high risk[41]. This model is under refinement and its ongoing development will be supported by the EPSRC AIDPitSurg grant (see Related Funding). All models will likely continually improve over time, with increasing refinements and a growing dataset.

Planned data collection for IDEAL Stage 1 and 2a studies

Feasibility and acceptability will be assessed via Acceptability of Intervention Measure and Feasibility of Intervention Measure questionnaires (WP1/2/3)[33]. Human factors analysis will include independent observation and qualitative interview, with any safety concerns analysed using the Systems Engineering Initiative for Patient Safety model (WP1/2/3)[42]. Educational yield questionnaires will be derived from the Intercollegiate Surgical Curriculum Programme workplace-based assessment proforma (WP2). Perceived clinical effectiveness and safety questionnaires (WP3) will developed specifically for this study. Operative performance based on video review will be assessed via a modified Objective structured assessment of technical skill (OSAT) scoring (WP3)[37]. Inpatient outcomes (WP3) will be collected via our internal prospectively developed database, which aligns with a previous multicentre collaborative study on patient outcomes, and includes: length of stay, hypernatraemia (including diabetes insipidus), hyponatraemia (including syndrome of inappropriate anti-diuretic hormone), new hormone deficit, cerebrospinal fluid leak and visual loss[14, 36]. In the future, we will add longer-term outcomes (such as recurrence, biochemical remission and quality of life) as our dataset becomes more longitudinal and multimodal (e.g. includes radiomics), but this is beyond the scope of the current proposal. All quantitative data will be descriptively presented, and qualitative data will be thematically analysed. Definite comparative statistics will not be performed during these IDEAL Stage 1 and 2a studies.

Study size

The chosen sample size for IDEAL Stage 1 studies (n=6) and IDEAL Stage 2a (n=30) is derived from the IDEAL Guidelines[30]. The intent of these studies is rapid iterative modification (of the AI-human interface), and preparation of the technology for future multicentre and definite comparative studies.

Resources and cost justification (1000 words)

Item	Cost (GBP)	Frequency	Total over 3 years
Data annotation	4000	Yearly	12000
Data storage	500	Once	500
Personal computer	2500	Once	2500
Scientific meetings	5000	Yearly	15000
Overseas training visit	3000	Once	3000

Data management
Core to any AI technology is data curation, storage and management. To create this preclinical video AI workflow model, we had to generate videos annotated with steps and instruments. The accuracy of this annotation is crucial as it provides the ground truth for these supervised learning models. However, this is time-consuming and requires technical expertise. We have established relations with 3rd party professional medical data labellers (Anolytics), who have provided consistent, high-quality, high-throughput labelling for our dataset. During this proposal, we aim to translate these AI models and perform early clinical evaluation.

In each work package, we will need to assess the accuracy of the candidate AI model, alongside the above study metrics. To do this, we must provide ground truth labelling for the video cases that the AI workflow model will be analysing. This amounts to 36 videos (6 for IDEAL stage 1, 30 for IDEAL stage 2) per WP, resulting in 108 total videos. Our mean video length is each video is 78.5 minutes (i.e. 4710 seconds) in length, resulting in 4710 frames to be annotated per video, when sampled at 1 frame/second. The total number of frames for annotation is 508,680 (4710 x 108). We have been quoted a price of 0.02 USD per frame from Anolytics, therefore the cost of ground truth annotations here is 10,174 USD. In addition, one of the outcomes of WP2 is to compare the change of performance (using the validated OSAT scoring system) in an AI intervention cohort (n=36; 6 for IDEAL stage 1 and 30 for IDEAL stage 2) and a retrospective control cohort (n=30). This means 66 videos must be annotated for OSAT scores, which we have elected should be done independently by expert annotators (and their scores averaged). We have received a quote of 50 USD per video from Anolytics for this, and have been satisfied with their annotation expertise for this task on previous projects. This results in an additional 3300 USD (66 x 50). Therefore, in total 13474 USD will be required for data annotation, which is approximately 12,000 GBP (or 4000 GBP per year across 3 years) after a 10% buffer is applied to contemporary exchange rates. I have opportunistically attached a quote to the head of department letters section.

From a clinician's point of view, the annotation of data can be quite time-consuming and can be technically challenging when using advanced labelling software. Thus, the advantages of using this service not only ensures the accuracy and validity of our ML models but will also allow significantly more time to be spent on other parts of the project (e.g. patient recruitment and technological iteration), improving project feasibility and scalability. Therefore, data annotation funding is a major allocation of my proposed yearly budget.

Furthermore, our operative videos are 4k quality, and on average are 50 gigabytes in size. Therefore, for the 138 videos above (WP1 36 videos, WP2 66 videos, WP3 36 videos), we will require 6,900GB of storage. We will store this data on cloud-based and physical encrypted hard drives. The former is provided by the Touch Surgery™ Enterprise platform (Medtronic), but I would need funding for the latter – for mobile encrypted solid state drives (SSD) to extract videos from endoscope video recorders and act as backup storage. In our experience with UCL Marketplace, such an SSD, with an associated USB-capable case and protective cover, can be purchased for approximately 500 GBP.

Finally, I would require a personal computer with the processing capability to manage large video datasets (editing, pre-processing, transfer) and basic statistical and machine learning analysis. For

example, a MacBook Pro or AlienWare laptop, which are generally between 2000-2500 GBP in cost.

Research dissemination
Participation in conferences is essential for disseminating research findings, receiving feedback, networking with peers and establishing collaborations. Therefore, a budget has been allocated for attending two major conferences relevant to the project per year. One of these would be a clinical conference, for example, the European Association of Neurosurgeons Congress or the World Federation of Neurosurgical Societies Congress. The other conference would be an AI or data science conference, for example, the International Conference on Medical Image Computing and Computer-assisted Intervention (MICCAI) or the Digital Neurosurgery Conference. I have budgeted up to 2500 GBP for each trip, to showcase the findings of the project and learn from international communities with shared interests.

Overseas Centre Training fellowship
In addition to the local training gained through UCL's existing courses and through learning in practice during this PhD, I would also like to gain an alternative experience via an external overseas centre. I have spent the last four years working within my current research group, and whilst this has been very productive and enjoyable, I believe I would benefit hugely from the opportunity to broaden my perspectives and spend a short period, for example, a 4-6 week research fellowship, in another centre. I have identified potential hosts at George Washington University, whom we collaborate with, and share interests in neurosurgical AI. Namely, Kevin Cleary, Professor of Bioengineering and Daniel Donoho, neurosurgeon and founder of the Surgical Data Science Collective and the Digital Neurosurgery Conference. They have worked together to perform multicentre AI work in the USA, and as multicentre work is the natural next step of my proposal's AI pipeline, I believe learning from their expertise would be particularly valuable to my ongoing development. I have received a letter of invitation and support from Kevin Cleary to support this training visit (uploaded to the head of department letters section).

3. Career Progression (700 words)
I aim to be a high-quality academic neurosurgeon, with an emphasis on advancing surgical technology and innovation. In pursuit of this, I have completed an Academic Foundation Programme in Clinical Neuroscience at the University of Cambridge, and a Clinical Research Fellowship in AI and Neurosurgery at University College London. I am now a run-through neurosurgical trainee with protected academic research time supported by an NIHR Academic Clinical Fellowship. The natural next step in my development as a clinician-scientist is completing my PhD, to further develop my general academic skills and technical skills (i.e. data science).

This proposal epitomizes my interests and ambitions, and having the opportunity to pursue it will give me the time, practical experience and formal training to develop these further. The majority of my research thus far has focussed on pre-clinical technology development of new technologies (robotics & AI) or clinical studies of established treatments and technologies. During this PhD, a key learning outcome for me is learning how to bridge this gap systematically and safely through early clinical translation. I hope to acquire the necessary skills in practice and via the learning opportunities available at UCL. For example, the ACCELERATE Potential and ACCELERATE Expertise courses (via the Academic Careers Office) to improve my translational research skills. Working closely with key industry collaborators is a core tenant of technological translation, and I will also build these skills by collaborating with two of the world's largest medical AI companies (Medtronic and NVIDIA). Furthermore, I plan on visiting an external centre in the USA for a short period (e.g. 4-6 weeks), to learn specific skills in multicentre collaborative AI, which is the next translational step after the work of my proposed fellowship (see Cost Justification section).

Similarly, this proposal will allow me to further my technical skills in data science and clinical AI. Again, I plan to do this in practice, and via courses offered by my institution. For example, the Clinician Coders Workshop (via the Academic Careers Office), to improve my technical data science skills. I also plan to apply to the annual Surgical Data Science Summer School in Strasbourg, with the support of my supervisors, both of whom have been faculty on this renowned data science programme. Furthermore, AI is a rapidly evolving space with near-constant advances in model development. Whilst I have basic coding skills, acquired through experiences such as R statistical software courses via the University of Cambridge, I cannot independently build AI models, and therefore I collaborate closely with AI engineers within our research group at WEISS. I think developing and maintaining this ability in light of the rapidly advancing nature of the field is difficult to achieve whilst balancing clinical, surgical and academic training. Therefore, my approach thus far has been to ensure I have enough practical domain knowledge to communicate effectively within our interdisciplinary team and critically appraise our work. I have done this through online learning platforms (FAST.AI) with the support of my AI engineering colleagues, and I hope to develop these skills further during this fellowship.

Furthermore, through the UCL Doctoral Skills Training Programme, I plan to bolster my wider academic skills in research management, data governance and scientific engagement. I plan to enrol in the UCL Healthcare Engineering Impact Fellowship if time allows, to bolster my skills in leadership, scientific communication, and patient and public engagement. Additionally, I am a faculty member of the UCL AI in Neurosurgery course, and will continue to deliver this course to maintain and develop my teaching skills.

Finally, after my doctoral studies, I plan to apply for an academic clinical lectureship and then academic consultancy upon completion of my training. I intend to subspecialise in pituitary surgery - to align my clinical and academic practice, and allow efficient and bidirectional bench-bedside innovation. This synergy will allow me to keep a patient outcomes centred approach to innovation, and drive further translation of the proposed AI technology to meet this goal in the future. This fellowship therefore optimally integrates my surgical and research interests, and will provide the ideal training for me to grow as a cutting-edge surgeon-scientist.

5. Applicant Capability to Deliver (1000 words)

Contributions to the generation of new ideas, tools, methodologies, or knowledge
My interests lie in applying technology and innovation to address neurosurgical problems. I have developed these interests through undergraduate research electives and post-graduate integrated clinical academic training. This has culminated in 65 peer-reviewed publications, 54 presentations (including at an international level), numerous academic awards and multiple grants.

My passion for innovation was first inspired in 2016 during a research elective at the Hamlyn Centre, Imperial College London. The focus of this 2-month project was first producing a literature review of the clinical needs in endoscopic neurosurgery, followed by a technological gap analysis to identify needs unmet by current technological solutions. I worked with engineers to generate a concept for the "NEUROCyclops", a novel robotic platform which used a parallel tendon manipulator within a tubular retractor system. We then generated a prototype, and executed a comparative preclinical validation study, presenting the findings internationally via a clinical and engineering conference. This project was incredibly educational. It was my first exposure to the principles of biological design and interdisciplinary collaboration, and allowed me to discover a creative outlet which I have exercised since.

I developed these skills further during my Academic Foundation Programme (Clinical Neurosciences theme) at the University of Cambridge. I applied novel robotic transcranial Doppler technology with finger plethysmography to noninvasively assess cerebral autoregulation in patients with post-concussive syndrome. I investigated the unmet need for this through a published review and then translated this technology via a first-in-human pilot study. I developed the protocol, attained ethical approval, collaborated with brain physicists and learned how to use specialised analysis software (ICM+) to execute this project. Leading this from concept to publication and international presentation was an important step in my academic development and technology translation experience.

I then joined WEISS UCL as a clinical research fellow in AI and neurosurgery, where I developed an AI model capable of workflow recognition for endoscopic pituitary surgery in collaboration with AI engineers and industry. This was the first of its kind in pituitary surgery, and I then attained an NIHR Academic Clinical Fellowship and Cancer Research UK Predoctoral Fellowship to develop these AI workflow models further, for example.

Through needs analysis, targeted innovation, and systematic evaluation I have gained a skill set in surgical technology and innovation, whilst contributing to the wider technological evidence base. Thus, I believe I have the necessary skill set to deliver this application.

Development of other researchers and maintenance of effective working relationships
Further to my surgical innovation interests, I have parallel passions for collaborative research and teaching. I was a junior principal investigator for a multicentre observational pituitary surgery outcomes study, recruiting almost every neurosurgical centre in the UK & Ireland. I executed this collaborative study while I held a national leadership position (Vice Chair of the Neurology and Neurosurgery Interest Group), and leveraged this to motivate, coordinate and collaborate with a network of >100 consultants, trainees and students. This study recruited >800 patients, representing the largest study of its kind, leading to multiple publications and presentations (for example). For this, I was awarded the prestigious Hugh Cairns Prize by the Society of British Neurosurgeons. It has created an effective pituitary research network, which I continue to manage and remains productive. I have applied these skills in teamworking across numerous other international projects, including with The Pituitary Society, therefore establishing myself within relevant national and international pituitary research networks, which I hope to leverage for future multicentre translation of my work.

I have received formal teaching training via the Teaching the Trainers Course (Royal College of Surgeons in England), and have attained an Associate Fellowship from the Higher Education Academy after being a clinical supervisor (Peterhouse and Queen's College, University of Cambridge) and completing the Integrated Foundation in Medical Education programme (University of Cambridge). I have applied these skills through numerous avenues since. For example, as a Trainee Lead at the Queen Square Neurosurgery Simulation Centre, where I have organised consecutive pituitary surgery simulation courses (receiving outstanding feedback, and

demand for a 4th edition of the course) and have co-led simulation research. I am also the Skull Base Trainee Lead of the Queen Square Annual Review Course, leading the coordination of teaching sessions on the contemporary clinical and academic updates within pituitary surgery – attracting excellent feedback from international delegates for 3 consecutive years. I hope to apply these teaching skills to improve my research engagement skills and my supervision skills as I develop as an academic.

Contributions to the wider research and innovation community

I am engaged in numerous initiatives to engage with the wider research community. I have co-led weekly academic "tweetorials" of contemporary pituitary research via the British Neuroendoscopy Society Twitter for two years. Similarly, I have produced critical analysis summaries of pituitary research via FacultyOpinions.org, and was awarded an Associate Faculty Award for my work. I have also engaged with the innovation community as a faculty member of the UCL AI in Neurosurgery Course, where I taught about critical appraisal of clinical AI research to an international audience, receiving excellent feedback. Finally, I am passionate about open science, and have curated 25 open-access annotated pituitary surgery videos as part of the International Endoscopic Vision Challenge.

Contributions to broader research or innovation, users and audiences, and towards wider societal benefit

I recognise the importance, of public and patient engagement, particularly for topical research such as AI, where I have studied wider perspectives on the technology. Applying this knowledge, I became faculty for the UCL SURGE project, where I led the creation of an animated video on my AI research, making my work and vision accessible and relatable. More recently, I have received funding (see Related Funding section) to co-lead international consensus studies on patient priorities and core outcomes in pituitary adenoma research. We have engaged with patients through societies and accessible videos (that I co-created) so that on completion, this work will form part of a research translation toolkit focussed on high-priority clinical needs.

Degree registration (250 words)
Have you registered for your PhD?
No, I have not yet registered for a PhD. I intend on registering if successful in attaining this fellowship. I have applied for time out of programme from my neurosurgical training in advance, and have discussed this with my training programme director.

Clinical activities and training (500 words)

Professional Registration Number	7647204
Clinical speciality or profession	Neurosurgery
Current training stage or nearest equivalent	ST3
Completion of training date	My listed CCT date is 08/2029, which will be extended to 08/2032 if I am successful in attaining a PhD fellowship.
Royal College Examination, with name of exam and date or anticipated date	- Completed: MRCS exams Part A (88%, 09/2018) and Part B (86%, 05/2019). - Upcoming: FRCS, post-PhD, during ST5-7.
Do you intend to be clinically active during or after the fellowship	Yes
Will you seek an honorary clinical contract	Yes, at the National Hospital for Neurology & Neurosurgery.
The percentage of your time you will spend on clinical duties	2 clinical sessions per week (i.e. 20%), please see below.
% of this time that will have direct relevance to your proposed work	See below.

I am currently a neurosurgery ST3 within the London deanery and an Academic Clinical Fellow at University College London. I entered-run through training in 2021, after receiving the Barrow Foundation UK Junior Neurosurgical Trainee Award for ranking 1st in national neurosurgical selection. Before training, I attained neurosurgical experience as a medical student, through three month-long sub-internships at international centres of excellence: Johns Hopkins (Baltimore, USA), Mayo Clinic (Rochester, USA) and Emory (Atlanta, USA) – all facilitated through competitive elective grants and completion of my USMLE exams. I then completed the Clinical Neurosciences Academic Foundation Programme at the University of Cambridge, during which I had neurosurgery, neurology and neurocritical care rotations. Following this, I attained further additional neurosurgical experience during a one-year Clinical Research Fellowship (50% clinical, 50% academic) at WEISS UCL.

Resultantly, I have been able to achieve the required clinical competencies for my stage of training throughout my ST1 and ST2 years, whilst balancing my academic duties – reflected by my ARCP and multi-consultant report outcomes. During ST1 and 2 at the National Hospital for Neurology & Neurosurgery (NHNN), I was afforded the opportunity to be on the registrar rota during my neurosurgical rotations, paired with a senior trainee. Therefore I have benefitted from two years of dedicated operating lists, supervised on-call shifts, clinics and MDT participation – ahead of my proper registrar years (ST3+). My goal for the end of ST3 is to be “emergency competent” such that I can safely carry out first-tier on-call registrar duties. I am confident I will be able to achieve this, given my clinical track record thus far.

During my proposed fellowship, I intend on spending 2 clinical sessions per week (20%) for the primary purposes of maintaining my surgical and wider clinical skills. This would be in the form of on-call shifts and theatre lists at NHNN. For on-calls, I would take these on an ad-hoc basis, to allow flexibility around my academic work. For theatre lists, I will aim to join elective pituitary surgery lists weekly – for training and for implementation of the proposed studies.

Mentors (500 words)
I have established academic and clinical mentors, whose mentorship I have benefitted from over the last few years, and I hope to continue to benefit from.

Academically, I have been supervised by Mr Hani Marcus since 2016, when I was a visiting third-year medical student from Dublin and he supervised me during a research elective at the Hamlyn Centre, Imperial College London. We executed and published a preclinical validation study of a robotic platform for use in endoscopic neurosurgery. This was an incredibly formative placement for me - igniting my passion for surgical technology and neurosurgery. Following this, Hani has mentored me over the last 7 years, and has been my formal academic supervisor for 3 years at UCL, during which we have published 37 peer-reviewed papers together (representing the majority of my academic output). This is a testament to our working relationship, and his supervision skills. We have achieved this through quarterly academic supervision meetings, monthly small-group AI research meetings, and weekly large-group research meetings. I plan to continue to this tried and tested schedule during my proposed PhD. I have other academic mentors (who are not supervisors on this project) whom I could organise to mentor me if the panel prefers, but reflecting on the above, it is my first preference to have him both as a mentor and supervisor on this project if possible - he is my role model.

Similarly, I have been supervised by Mr Laurence Watkins since 2021, when I entered run-through neurosurgical training. He is the London Neurosurgery Training Programme Director and a Senior Consultant Neurosurgeon at NHNN. He was my clinical supervisor for my first neurosurgical rotation during training, and has guided me since during my surgical training. We have had scheduled clinical supervision and training meetings at least bi-annually, and he has always been available ad-hoc for training and pastoral support during this time. Resultantly, I have been able to excel in my clinical training thus far, as demonstrated by my multi-consultant feedback reports, multidisciplinary team feedback assessments, surgical logbook numbers and ARCP reports. During this proposed fellowship, I would plan to meet with him 1-2 yearly, to ensure support for the maintenance of my clinical skills and support my eventual return to clinical training after my PhD. I plan to use the Intercollegiate Surgical Curriculum Programme Objective-Setting Supervision Meeting template (as used during my previous clinical rotations), to ensure supervision is structured and targeted to all domains of my clinical development.

Head of department statement
Online

Supervisor statements (1000 words per supervisor)
Online

Optional: Project partners (1000 words)
Provide details of any project partners' contributions, and letters or emails of support from each named partner.

Project partner name	**Project partner address** Postal address	**Country**	**Direct contribution**	**Direct contribution value**	**In-kind contribution**	**In-kind contribution value** Provide the estimated value in £s	**Name of contact***	**Job title**	**Department**
Medtronic	Building 9 Croxley Park Watford, Herts WD18 8WW	UK	/	/	Technical Support	£50000	Karen Kerr	Research director	Research

My proposed fellowship involves industry collaborators (please see the Industry Collaboration Frameworks section below). I have also attained a letter of support from Prof Peter McCulloch, Chair of the IDEAL Collaboration and Professor of Surgical Science and Practice at Oxford University, as an academic collaborator whom I intend to seek ad-hoc review of my work this fellowship. He is not a project partner by definition so I have not included him in the table above.

Optional: Industry Collaboration Framework (1500 words)
The project is supported by a global leader in medical technology, services and solutions, Medtronic. Medtronic (via Touch SurgeryTM Enterprise) provide a world-leading secure data management and annotation platform, allowing us to continually develop our state-of-the-art AI models. They have provided a letter of support as a supplier for this application (uploaded in the head of department letters section).

We will also work closely with the world leader in AI Computing, NVIDIA, who have supplied us with a Clara AGXTMplatform and their technical support as part of a related EPSRC grant (see Related funding section). This technology allows state-of-the-art efficient embedded computing systems with display monitors, allowing us to store AI models and present workflow analysis outputs with clinicians.

ICF forms were not completed as neither company is expecting to access the outputs of the research prior to publication.

Facilities (250 words)
The proposed clinical studies will be performed in the National Hospital for Neurology and Neurosurgery. Touch Surgery™ Enterprise is already in use within the operating theatres there, with the necessary governance and biomedical approvals in place. These approvals are in process for the NVIDIA Clara AGX™ system. Operations will be performed as per routine clinical care, and any patient or staff participants in our studies will only do so with informed consent. These studies will not draw from any additional clinical resources or cost to the hospital.

These studies will be supported by technical expertise from WEISS, a state-of-the-art facility equipped with cutting-edge resources such as computation infrastructure (high-performance computing clusters, dedicated NVIDIA DGX station and DGX-A100). As a centre, WEISS offers a robust infrastructure for clinical translation, including a dedicated team supporting research ethics, clinical trials, quality management and public engagement. Any engineering or technical demands of this proposal (i.e. iterating AI models between IDEAL stages) will be supported by an existing EPSRC AIDPitSurg grant (see Related Funding section).

Data management and sharing (1500 words)

Proposal name
Towards AI-Assisted Pituitary Surgery – The Systematic Translation of Surgical Workflow Analysis to Improve Training, Audit and Prognostication
1.Data description
1.1.Type of study This is an interdisciplinary project that aims at clinical translating artificial intelligence (AI) driven surgical workflow analysis in pituitary surgery for the purposes of audit, training, and prognostication. Data from patients undergoing transsphenoidal pituitary surgery are required to train the AI techniques, as well as user feedback data from health care professionals using the technology. **1.2.Types of data** The collected data will be multimodal, including intraoperative surgical videos (with annotations for workflow & performance) and postoperative outcomes. Patient demographic data (age, sex, tumour type) will be collected to give context for publications. User data from healthcare professionals will include feedback questionnaires, including acceptability, feasibility, and usability. **1.3.Format and scale of the data** Data from up to 108 patients undergoing pituitary surgery is collected. Patient's demographic and outcomes data will be in XLS format. Video data (without audio) will be in MPG format. Annotations will be stored as a separate file and will be in XLS format. User feedback data will be stored as XLS format. All these are the standard formats for this type of data and enables sharing and long-term validity of data.
2.Data collection / generation

2.1.Methodologies for data collection / generation

A data collection pipeline is already built, established via a Cancer Research UK Predoctoral Fellowship and an NIHR Academic Fellowship, with ethics approval (IRAS 271696; REC Ref 21/SW/0027) and local data governance approval. Patient data is generated as part of the standard clinical practice already. User data and annotation data are the only new data types being generated for these projects.

Patient's demographic and outcomes data are stored via an existing database in XLS format on the NHS secure National Hospital for Neurology and Neurosurgery (NHNN) network drive. This data will be backed up in a codified pseudonymised fashion (see below) on an encrypted solid state drive (SSD) which will be kept locked on hospital premises, in the lead supervisor's clinical office.

Intraoperative videos are collected using the secure Touch Surgery™ Enterprise platform by Medtronic. The platform fully anonymizes the videos before storing (redacting any footage which is extranasal) and making them available on the secure cloud based server (region-specific, GDPR compliant, Amazon Web Server). This data will be backed up on the same encrypted SSD which will be kept locked on hospital premises, in the lead supervisor's clinical office.

Anonymised video annotation exports and user data will be stored on an encrypted SSD, and backed up via UCL Data Safe Haven if required, which offers a secure solution for data storage, handling and analysis.

2.2.Data quality and standards

The quality and consistency of the multimodal data is ensured by documenting patient's complete record. Patient's record (demographics and outcomes) are collected by a senior clinical fellow as part of standard care, following guidelines from a multicentre pituitary surgery outcomes study led by our group (10.3389/fonc.2022.1049627) to ensure completeness and accuracy. Each record will be given a codified number, to allow linking to the related video, without sharing of any identifiable information.

Surgical videos are recorded from the same HD Storz Karl Endoscope across different cases. The video data is fully anonymized at the point of recording as mentioned above. No identifiable patient or user data will be stored. Each video will be given a codified number, to allow linking to the related demographics and outcomes data. Similarly, each user feedback data point will be given a codified number to link to the study case it pertains to.

3.Data management

3.1.Managing, storing and curating data.
As the patient and video data is collected routinely for clinical purposes, it will continue to be stored on NHS machines indefinitely as part of the patient's clinical care. The master clinical database (demographics and outcomes data) is protected by encryption and passwords, and access is restricted to the clinical research team. As above, a codified project-specific ID number is assigned to each patient who consents to the use of their data for research and is included in the database.

Pseudonymized data will be stored will be backed up on an encrypted SSD (which will be kept locked on hospital premises, in the lead supervisor's clinical office) or in the UCL Data Safe Haven, depending on the data type. Only fully anonymized video and annotation data is shared with the engineering team for AI model development and evaluation, and with data annotation services (via the Touch Surgery platform).

3.2.Metadata standards and data documentation

Study protocols will be published *a priori* in open-access repositories where possible, and/or will be published with any publications resulting from these projects. Annotation protocols, and experimental code will be stored and actively maintained on the WEISS Open Research platform.

3.3.Data preservation strategy and standards
The pseudonymized data (video, annotation, and user data) will be stored in an encrypted SSD with access to only authorized persons and/or the secure UCL Data Safe Haven and/or Touch Surgery Enterprise™ for a minimum of 10 years. As per our Medtronic research contract, we retain full and exclusive ownership rights to uploaded video data as the research team. Any other identifiable data such as consent formed will be stored in a locked filing cabinet at NHNN for 10 years after study completion.

4.Data security and confidentiality

4.1.Formal information/data security standards

Data will be stored as per ICH-GCP and GDPR data protection principles (ICO reference Z6364106).

4.2.Main risks to data security

The risk of theft or loss of confidential data is low. Data will be stored in accordance with UCL data security policy, and NIHR guidance on data security. All data stored on secure UCL servers and/or the secure SSD will be pseudo- or fully anonymized. Participants will be identified by codified project specific ID numbers, that are linked by key to identifiable clinical data stored separately and securely within NHNN. Identifiable/sensitive information will not be shared with or given to anyone except research sponsors, regulatory authorities, and primary researchers. Data sharing will only occur where consent has been explicitly obtained from participants, with all identifiable information removed.

5.Data sharing

5.1.Suitability for sharing
Fully anonymized data is a valuable resource for advancing research in this area, and is suitable for sharing. This data will be made publicly available along with the respective publication to benefit the research community, after consent is sought from any participants (patient and user).

5.2.Discovery by potential users of the research data
Users may find out about the availability and existence of our data through information posted on the study website, journal publications, and updates on our laboratory twitter page. Our policy for data sharing will be published on the study website and described in publications. We are fully committed to making data available to interested parties. The fully anonymized data will be made publicly available on the WEISS Open Data Server along with the respective publication along with the license and policy for data sharing.

5.3.Governance of access
The PI and supervisors will ask potential data users to provide a brief research proposal detailing how they wish to use the data and types of data requested. This may form part of a data sharing agreement. Consent will be obtained from participants prior to sharing of fully anonymized datasets. Data and software will be shared via the UCL WEISS Open Research platform and independent oversight will be sought from the WEISS executive committee.

5.4.The study team's exclusive use of the data
The fully anonymized data and any associated software will be shared along with the respective paper publication.

5.5.Restrictions or delays to sharing, with planned actions to limit such restrictions
Informed written consent will be sought from participants for sharing fully anonymized data prior to study participation. Participants will be made aware of the data sharing policies during consent. There are no other delays or restrictions.

5.6.Regulation of responsibilities of users
To ensure that data are appropriately used, a data sharing agreement that sets out criteria for data access and conditions for use maybe required. This will include privacy and confidentiality standards, as well as citation and publication rights as per the NIHR Data Sharing Policy. In some instances of research collaboration using shared datasets, co-authorship on publications could be more appropriate. It would also be expected that researchers using shared data acknowledge funding support from the NIHR for the original study.

6.Responsibilities

The project team (fellow, supervisors, and collaborating AI engineers) will be responsible for the data management, metadata creation, data security and quality assurance of data. Some data will be hosted on Medtronic, NHNN and UCL servers, hence the Medtronic, NHNN and UCL IT services will also be responsible for data security.

7.Relevant policies on data sharing and data security

Data Management Policy	https://www.ucl.ac.uk/library/open-science-research- support/ research-data-management/data-policies-and-expectations
Data Security Policy	https://www.ucl.ac.uk/isd/sites/isd/files/ucl_research_data_policy_v6.pdf https://www.ucl.ac.uk/isd/services/file-storage-sharing/data-safe-haven-dsh
Data Sharing Policy	https://www.ucl.ac.uk/library/open-science-research- support/ research-data-management/best-practices/how- guides/sharing-data

Institutional Information Policy	https://www.ucl.ac.uk/information-security/information-security-policy https://www.ucl.ac.uk/interventional-surgical-sciences/weiss-open-research
8. Author of this Plan	
Danyal Z Khan 07427237473 d.khan@ucl.ac.uk	

Ethics and responsible research and innovation (500 words)
The principal issues potentially associated with this work pertain to data management and security, and AI intervention implementation in each work package. The former is covered extensively in the Data Management and Sharing section, with reference to security measures and information standards. This is supported by existing ethical approval (IRAS 271696, REC 21/SW/0027, UCL/UCLH as sponsors) and local data governance and IT approvals at NHNN. These approvals are currently focussed on WP2, and detail the collection of operative videos (via Touch Surgery™ Enterprise) & associated clinical data, development of AI models, display of workflow analytics to the surgical team, and collection of surgical team feedback (including acceptability, feasibility and usability). Any workflow analytics displayed intra-operatively would not be displayed to the operating surgeon, therefore the risk of distraction is minimised. Therefore, WP2 does not need any further amendments and we are satisfied any potential ethical issues have been addressed.

WP1 and WP3 would require minor amendments to our existing ethics. For WP1, this would include the display of AI-generated operative notes to assist surgeons in operative note completion. The risk here is low, as there is a human in the loop (the surgeon) who will review any AI output, and will be responsible for any amendments to the note to reflect clinical events. This approximates existing practice at NHNN, where we often use templated operative notes within our electronic medical record system that surgeons amend as appropriate, and therefore we anticipate no/minimal additional risk with AI integration. Similarly, for WP3, we will display a post-operative complication risk profile to clinicians, but again there is a human in the loop, and any management decisions will ultimately be made by the responsible senior decision maker for the patient. Therefore, we anticipate no/minimal additional risk with AI integration.

Finally, with regard to data diversity, NHNN is the UK's largest pituitary centre, with an international patient base. All consent forms and processes will be available in non-English languages as necessary, and adaptations will be made to maximise inclusiveness if possible (e.g. audio narration for those who are visually impaired). I have also engaged in numerous public and patient engagement initiatives (see Applicant Suitability to Deliver section) to maximise accessibility to our work. Therefore, we hope our work is inclusive, and our study dataset is diverse and representative of a wide array of populations.

Research involving human participation (700 words)
Each WP will involve patients with pituitary adenomas, undergoing endoscopic endonasal transsphenoidal surgery at NHNN. Each patient will be consented by a study team member before their planned operation. Participation will be voluntary and will not affect clinical care. Similarly, surgical team members will be invited to provide user feedback voluntarily and will be consented by a study team member if they consent to inclusion.

In the Ethics and Responsible Research and Innovation section, we have detailed the approvals in place, anticipated issues and plan for ethical amendment. Here, we have also detailed our plans for improving the inclusiveness and diversity of our studies.

In the Reproducibility section, we have detailed sample size justification and the data to be collected. In the Data Management & Sharing section, we have detailed how this data will be securely stored, managed and accessed.

Clinical research in the NHS, public health or social care (250 words)
This research will take place in an NHS setting. We already have ethical approval and HRA approval (as per the above sections), which simply requires minor amendments. There are no additional significant research costs, NHS support costs or NHS treatment costs. Any potential research costs (e.g. consenting patients, data management, administration) will be completed by myself as a research fellow. Our assistive technology is on loan, and we will receive technical support from industry suppliers. I have however completed a SoECAT online just in case (pending review by North Thames CRN) and have sought advice from my local AcoRD specialist. I will forward the outcome when available, as instructed during our email correspondence on 08/09/2023.

Related applications (500 words)
Is this application related to another application to NIHR or other funding organisation?

This application builds on my previous funding, and will be supported by allied funding within my group.

Personal funding
I have built the foundations of this PhD proposal primarily with the support of two previous personal fellowships, an NIHR Academic Clinical Fellowship (of which I am in the third and final year) and a Cancer Research UK Predoctoral Fellowship (awarded March 2022, for 20,000gbp over 1 year). These programs have allowed me to collect pilot data (annotated videos and associated clinical data) for 100 pituitary adenoma patients – representing the largest dataset of its kind. They have also supported the building of a pipeline for secure data collection (including regional ethics committee approval and industry collaboration), and the development of preclinical AI models capable of accurately and automatically recognising operative workflow[23-26]. Finally, I am also a co-investigator on a National Brain Appeal Small Acorns Fund grant and a WEISS Public Engagement grant to investigate patient priorities in pituitary adenoma research and develop a core outcomes set via consensus. These studies are ongoing, and will form part of a research translation toolkit for pituitary adenoma patients, so that we focus translation on high-priority clinical needs, and measure the impact of interventions in a standardised fashion https://twitter.com/PitCop2023.

Group funding
Our group has recently received an EPSRC project grant (EP/Y01958X/1) to optimise these preclinical AI models further (540,000gbp, over 18 months, starting in October 2022). My proposed PhD supervisors are co-investigators and the senior leads on this project, which will fund a clinical research fellow to collect and annotate surgical videos (currently ~250 videos) and two computer science research fellows to drive AI model improvement. This project will be at its halfway point and well-established by my proposed starting time (August 2023), dovetailing with my PhD. fellowship plan which plans to clinically translate these preclinical models. However, my proposal is not dependent on its success, as we already have ready-to-translate AI workflow models, as cited above. I am not listed as a co-investigator on this grant, and I am receiving no personal support from this funding.

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